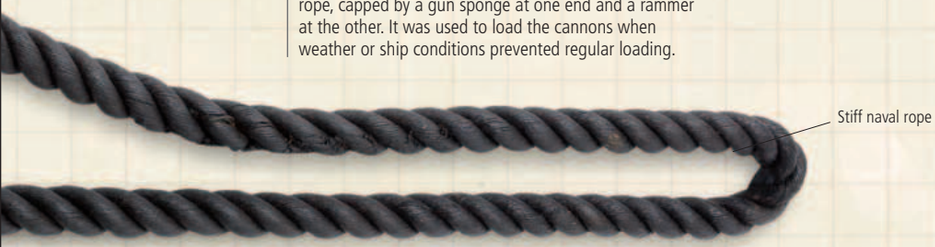
FLEXIBLE SPONGE AND RAMMER The flexible sponge and rammer consisted of a single piece of stiff naval rope, capped by a gun sponge at one end and a rammer at the other. It was used to load the cannons when weather or ship conditions prevented regular loading.

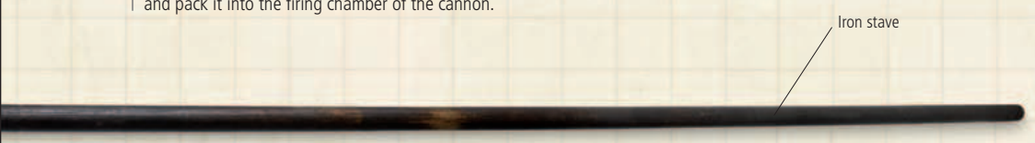




SPONGE This sheepskin sponge was kept damp and was pushed down the barrel of the cannon after every shot had been fired. This was to extinguish any residual burning embers, thereby preventing accidental discharges when the next cartridge was loaded.



RAMMER The rammer featured a wooden stave and concave ramming head, and was used to push the charge, wads, and shot down the full length of the bore and pack it into the firing chamber of the cannon.



WORM This double-piece iron corkscrew was used to extract wad and cartridge if a cannon had to be unloaded, or was used to remove flannel cartridge bases stuck in the bottom after the cannon had been fired.

SHELL TYPES

Naval ammunition generally had three purposes: punch holes in the sides of an enemy ship's hull; bring down the masts and sails; and kill enemy personnel. For these contrasting goals, various types of cannon shot were developed. Accuracy and range with anything other than round shot was generally poor, but these munitions were principally used when ships were in close quarters.



ROUND SHOT A single iron ball the size of the cannon's bore, used for its penetrating effect against the ship's thick wooden hull.

CHAIN SHOT Two or more cannon balls linked together—when fired, chain shot would scythe down enemy crew.

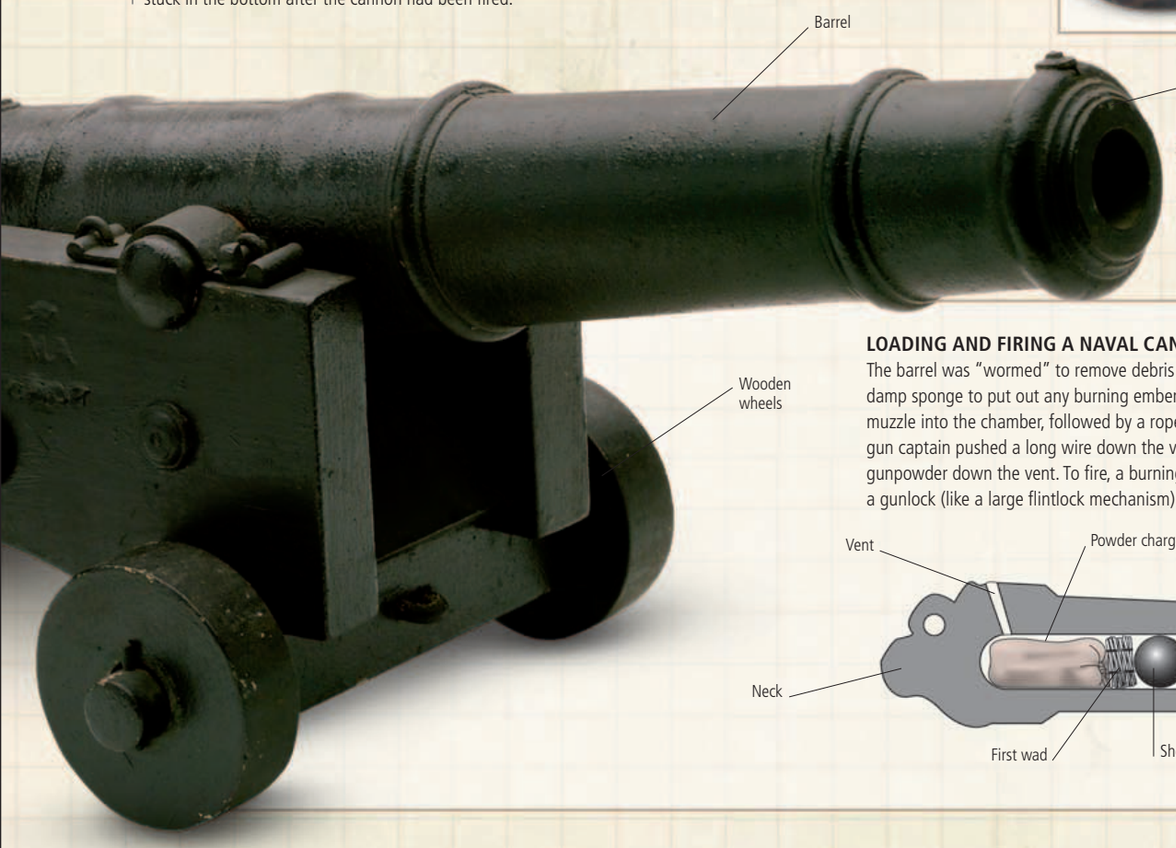




BAR SHOT Two or more pieces of shot linked by either a fixed bar or by extendable bar sections. Bar shot was designed to hack away at lines and rigging as it flew over the top of the ship's deck.



GRAPESHOT Balls of metal inside a tin or canvas bag. The container shattered when cannon were fired, producing a hideous shotgun-like effect against enemy crew.



24-POUNDER
This 24-pounder cannon is from the famous ship HMS *Victory*. She also carried 32- and 12-pounder cannon.

LOADING AND FIRING A NAVAL CANNON

The barrel was "wormed" to remove debris from the previous shot, then "swabbed" with a damp sponge to put out any burning embers. A fabric powder charge was loaded down the muzzle into the chamber, followed by a rope wad, then the shot, and then another wad. The gun captain pushed a long wire down the vent hole to prick open the charge, and poured gunpowder down the vent. To fire, a burning slow match was applied to the vent hole, or a gunlock (like a large flintlock mechanism) was triggered.

